

UBT203: BIOCHEMISTRY AND MICROBIOLOGY LAB

L T P Cr

0 0 4 2.0

Course objective: To provide provide fundamental understanding of protocols in basic biochemistry and microbiology with learning technique like TLC, UV-visible spectroscopy and microscopy. The course has an industrial reference as they form part and parcel of various SOPs related to microbiology and biochemical analysis.

Detail contents:

Microbiology:

Light Microscopy, microscopic examination of different microbes; Gram staining, spore staining; Preparation of culture media and sterilization techniques; sources of microbial contamination, isolation of heterotrophs and autotrophs; enumeration of microbial population in soil and water; microscopic measurement of cell dimension and growth by cell counting, growth measurement and growth curve.

Biochemistry:

Preparation of Buffers (Acidic/Basic/Neutral); Qualitative test for Carbohydrates, Fats, Proteins and Amino Acids (includes TLC separation also); Quantitatively estimate sugars by preparation of Standard curve of D-Glucose by DNSA method (includes principles of UV-Visible spectrophotometry); Standard curve of Protein by Folin Lowry's Method (including estimation of unknown), Determination of acid and saponification value of fats, Determination of optimal conditions (temperature, pH, Substrate concentration) for optimal activity of amylase enzyme, Separation and estimation of photosynthetic pigments (chl a & chl b) from spinach leaves (TLC/Spectrophotometric determination)

Course Learning Outcomes (CLO):

Students will be able to

1. analyze and distinguish different biomolecules which play a crucial role in process of life
2. analyze, compose and effectively communicate the observations and results of scientific investigation in a standard format.
3. describe some of the general methods used in the study of microorganisms.
4. demonstrate aseptic microbiological techniques in the laboratory and check sources of microbial contamination and their control.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Text books

1. K. Wilson and J. Walker (ed.), Practical Biochemistry, Principles and Techniques, Cambridge University Press, 1995
2. S.K. Sawhney and Randhir Singh (ed.) Introduction Practical Biochemistry, Narosa Publishing House, 2009
3. Pelczar Jr., M.J., Chan, E.C.S. and Krieg, Noel R., Microbiology, McGraw Hill (2003) 5th ed.
4. Microbiology- a Laboratory Manual, 2nd Ed, JG Cappuccino and N Sherman, Benjamin/Cummings Publ. Company, Inc. 1987.

Reference Books

1. R. Boyer, Modern Experimental Biochemistry, 3rd Ed., Pearson Education (Singapore) Pvt. Ltd., 2001
2. R. L. Switzer and L. F. Garritty, Experimental Biochemistry, 3rd Ed., W. H. Freeman, 1999
3. Tortora, G.J., Funke, B.R., and Case, C.L., Microbiology- An Introduction, Pearson Education (2007) 8th ed.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	Sessionals (May include assignments / quizzes / performance)	100

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